

WILD HUNT STORE

SHOP
SIMULATOR

A SIMPLE SHOP SIMULATOR PROGRAM

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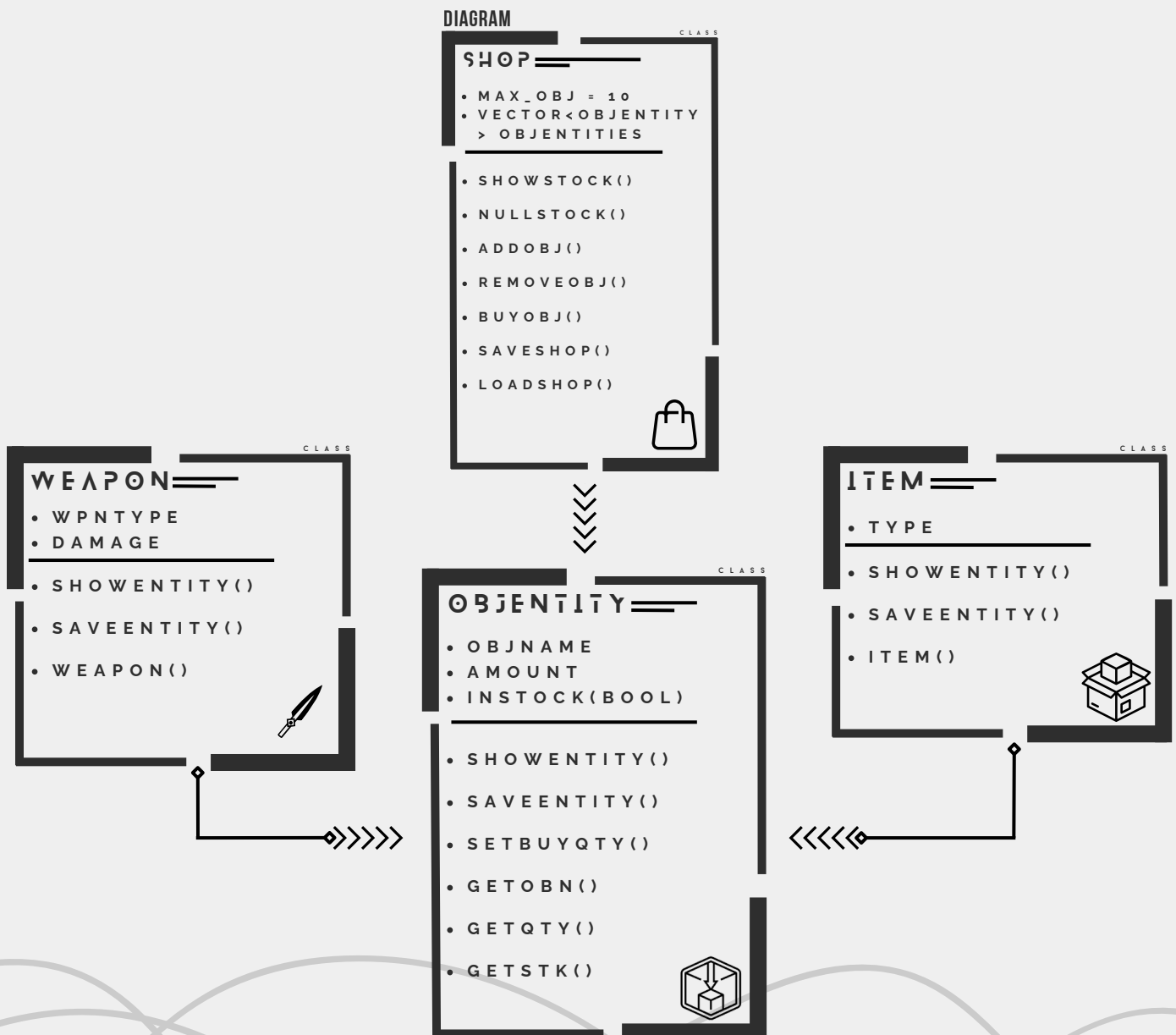
INTRODUCTION

MY PROGRAM IS A SIMPLE SHOP INVENTORY SIMULATOR WHERE THE USER CAN SELL ITEMS OF ANY TYPE, IT COULD BE POTIONS, MATERIALS OR WHATEVER ABOMINATION THE USER WANNA SELL, AND WEAPONS, A REALLY GOOD TYPE OF STUFF TO SELL, I GUESS?? (CUZ RPG ASHHAHAS).

IT CONTAINS A TOTAL OF 4 FUNCTIONS WHICH ARE

- SHOW STOCK (TO SHOW WHAT THE USER IS SELLING)
- ADD OBJ (TO ADD OBJECT TO SELL)
- REMOVE OBJ (TO REMOVE OBJECT CURRENTLY IN STOCK)
- BUY OBJ (TO REDUCE THE STOCK OF THE ITEM CURRENTLY IN STOCK

CLASS DIAGRAM



LOOP INTEGRATED IN THE PROGRAM

OOP IS SHORT FOR OBJECT-ORIENTED PROGRAMMING

ENCAPSULATION

APPLIED BY USING ACCESS SPECIFIER "PRIVATE" IN THE CLASS FOR VARIABLES. IT CANNOT BE ACCESSED FROM THE OUTSIDE, BUT IT CAN BE CONTROLLED AND ACCESSED THROUGH FUNCTIONS LIKE GETOBN() IN THE PUBLIC SECTION.

```
class ObjEntity {  
private://ENCAPSULATION  
    string ObjName;  
    int Amount;  
    bool InStock;  
  
public:  
    //GETTERS  
    string GetOBN() const { return ObjName; }  
    int GetQTY() const { return Amount; }  
    bool GetSTK() const { return InStock; }
```

INHERITANCE

THERE IS TWO TYPES OF OBJECTS IN THE PROGRAM, AND BOTH OF THEM ARE DERIVED FROM THE CLASS OBJENTITY, WHY? BECAUSE THEY ARE STILL AN OBJECT THAT INHERITS ALL OF WHAT THE CLASS OBJENTITY HAVE. THEY JUST HAVE THEIR OWN VALUES LIKE WEAPON DAMAGE FOR WEAPON.

```
class Weapon : public ObjEntity {  
private://ENCAPSULATION  
    string WPTYPE;  
    int Damage;
```

POLYMORPHISM

THIS IS APPLIED THROUGH VIRTUAL FUNCTIONS AND METHOD OVERRIDING. POLYMORPHISM IS USED HERE THROUGH THE SHOWENTITY() VIRTUAL FUNCTION. BY DEFINING IT IN THE OBJENTITY BASE CLASS AND OVERRIDING IT IN WEAPON AND ITEM. IT ENSURES BASE CLASS POINTERS AUTOMATICALLY TRIGGER THE CORRECT DERIVED-CLASS BEHAVIOR AT RUNTIME.

```
//POLYMORPHISM  
void ShowEntity() const override {  
    cout << "| Name: " << GetOBN() << endl  
    << "| Type: " << WPTYPE << endl  
    << "| Damage Stat: " << Damage << endl  
    << "| Stock: " << GetQTY() << endl;  
}
```

ABSTRACTION

THIS IS APPLIED BY USING A PURE VIRTUAL FUNCTION (SAVEENTITY) IN THE BASE CLASS OBJENTITY. IT ENFORCES A CONSISTENT INTERFACE WHILE HIDING INTERNAL LOGIC. WHICH MEANS EACH DERIVED CLASS MUST PROVIDE ITS OWN SPECIFIC IMPLEMENTATION. IT KEEPS THE STRUCTURE GOOD FOR SAVING AND HIDES INTERNAL DETAILS ALONG THE WAY.

```
//MEMBER FUNCTIONS  
virtual void ShowEntity() const = 0;  
virtual void SaveEntity(ofstream& out) const = 0;
```

WHO DID WHAT?

ME..... LITERALLY..... (TIRING ASF BUT FUN TO DO)